

Table 1: Benchmark results (F1 scores) across models and tasks. All models evaluated on 500 samples per task using the validation split with `max_ctx=2048` and `max_new_tokens=32`.

Model	SWDE	FDA	SQuAD	Avg.
GLA (baseline)	6.63	4.98	9.02	6.88
MS-GLA [1,2]	7.19	5.84	9.04	7.36
MS-GLA [1,2,4]	8.77	5.68	8.93	7.80
MS-GLA [1,2,4,8]	5.27	4.29	8.67	6.08

Table 2: Language modeling perplexity (lower is better) and downstream task accuracy (higher is better). All models use `max_ctx=2048`, `stride=1024`.

Model	Wiki. ppl ↓	LMB. ppl ↓	LMB. acc ↑	PIQA acc ↑	Hella. acc_norm ↑	Wino. acc ↑	Avg. PPL ↓	Avg. Score ↑
GLA (baseline)	20.02	26.54	17.31	63.87	32.93	51.62	23.28	41.43
MS-GLA [1,2]	19.79	23.24	19.04	65.56	33.83	50.67	21.52	42.27
MS-GLA [1,2,4]	18.85	21.89	19.29	64.74	33.53	51.22	20.37	42.20
MS-GLA [1,2,4,8]	19.86	21.91	19.29	63.71	33.56	49.57	20.89	41.53